Answer Key & Teacher Notes

Session 1 — Comparing Ratios with Common Terms · Session 2 — Comparing Rates with Benchmarks

HOW TO USE THIS LESSON'S PAGES

one lesson = two sessions

Warm-Up	About 10 minutes; use items 1–4 for a 5-minute check. Key: Unit Warm-Up Key.
Prerequisite Practice A/B	Before the lesson, for students who need the skill the session uses. Key in that folder.
Guided Notes	During the lesson. Students fill in the blanks as you teach; answers below.
Practice Form A	In class, after the lesson. Two pages. The self-check code tells students if they are on track.
Practice Form B	Same skills, new numbers: reteach, retake, small group or absent students.
Exit Ticket	Last 5 minutes. Two per page; cut in half.
Homework Packet	Sheet 1 after Session 1, Sheet 2 after Session 2. Level 1 has the same problems and answers, with a word bank, step frames and larger type for students who need language support.

SESSION 1 · TEACHER NOTES

Comparing Ratios with Common Terms

TALK PROMPTS

- Why does making the red paint equal (6 cups) make the comparison so simple?
- Both plant-food bottles use more water than plant food. Why does matching the drops settle the question?

PARTNER TASK

Orange Juice, Bubble Solution, Plant Food. One partner chooses the common multiple and scales the first mix; the other scales the second mix. Together, compare the other term and say which mix is stronger.

WATCH FOR

- Comparing second terms without matching the first terms ("7 > 5, so B is greener").
- Adding to match the first terms (turning 2 : 5 into 6 : 9) instead of multiplying.
- Thinking the mix with more water is the stronger mix.

SESSION 2 · TEACHER NOTES

Comparing Rates with Benchmarks

TALK PROMPTS

- Why is scaling to 30 tokens easier than dividing 7 by 3?
- Do the unit prices and the 12-bag totals point to the same store? Why should they agree?

PARTNER TASK

Gummy Bears, Typing Contest, Corner Store Chips. One partner compares with unit rates; the other compares with a benchmark. Check that both methods name the same winner, then switch methods.

WATCH FOR

- Subtracting the two quantities (9 − 2) instead of comparing rates.
- Choosing a benchmark that one amount cannot reach exactly (10 for a 4-pack and a 6-pack).
- Picking the larger total cost as the better buy instead of the lower cost for the same amount.

ITEM ANALYSIS

suggested DOK; judge the reasoning required, not the question label

Sheet	Standard	1	2	3	4	5	6	7	8	9	10	11
S1 · Form A	6.RP.A.3b	1	1	2	2	2	2	2	3	2	1	2
S1 · Form B	6.RP.A.3b	1	1	2	2	2	2	2	3	2	1	1
S2 · Form A	6.RP.A.3b	1	1	2	2	2	2	2	3	2	1	1
S2 · Form B	6.RP.A.3b	1	1	2	2	2	2	2	3	2	1	1

Items before the last three are Practice; then Find the Error (explain and correct) and two Test Practice items.

GUIDED NOTES

common multiple: A number both terms can be **scaled** to.

least common multiple: The **smallest** number that is a multiple of both numbers.

stronger: More of the main ingredient for the **same** amount of water.

- To compare two ratios, scale them so one term is the **same** in both.
- Then compare the **other** term. Less water for the same soap means a **stronger** mix.

Example: A common multiple of 2 and 3 is **6**. Mix A × 3 = 6 : **15**. Mix B × 2 = 6 : **14**. Compare the water: 14 cups is less than 15 cups. Answer: Mix **B** is stronger: less water for the same 6 spoons of soap.

YOU TRY

1. A: 6 : 27; B: 6 : 24. Tea B is stronger (less water).

2. a. 35 b. 24

EXIT TICKET

1. a. 15 b. 22

2. A: 6 : 9; B: 6 : 10. Mix A (less water)

3. **B** (less water)

PRACTICE FORM A

1. a. 32 b. 30 c. 27 d. 14

2. Mix A × 2 = 6 : 8; Mix B × 3 = 6 : 9. Mix B is bluer (more blue for 6 cups red).

3. a. Pitcher 1: 6 : 10; Pitcher 2: 6 : 9 b. Pitcher 2 (less water)

4. a. M: 20 : 36; N: 20 : 35 b. Bottle N (less water for 20 drops)

5. A: 6 : 18; B: 6 : 16. Cup B (less milk)

6. Pool 2: 6 tablets for 10,000 gallons, more than Pool 1's 5

7. Jar 1: 15 : 35; Jar 2: 15 : 36. Jar 1 is stronger (less water).

8. a. Team 1: 20 : 45; Team 2: 20 : 48 b. Team 1, because it needs fewer games (45) for the same 20 wins.

9. The white amounts are different, so 7 and 5 cannot be compared directly. At 6 cups white: A = 6 : 15, B = 6 : 14. Mix A is greener.

10. **B** (one term equal)

11. **A** (2 to 5)

Self-check word **COMMON**

PRACTICE FORM B

1. a. 24 b. 63 c. 15 d. 20

2. Mix A × 3 = 12 : 15; Mix B × 4 = 12 : 16. Mix B is bluer (more blue for 12 cups red).

3. a. Pitcher 1: 12 : 21; Pitcher 2: 12 : 20 b. Pitcher 2 (less water)

4. a. M: 15 : 40; N: 15 : 39 b. Bottle N (less water for 15 drops)

5. A: 6 : 20; B: 6 : 21. Cup A (less milk)

6. Pool 1: Pool 2 uses 6 tablets for 10,000 gallons, fewer than Pool 1's 7

7. Jar 1: 6 : 15; Jar 2: 6 : 16. Jar 1 is stronger (less water).

8. a. Team 1: 12 : 32; Team 2: 12 : 30 b. Team 2, because it needs fewer games (30) for the same 12 wins.

9. The red amounts are different, so 10 and 8 cannot be compared directly. At 12 cups red: A = 12 : 32, B = 12 : 30. Mix A is bluer.

10. **B** (12 : 32)

11. **B** (less milk)

Self-check word **EQUALS**

GUIDED NOTES

benchmark: A shared amount you scale **both** choices to.

rate: Compares two amounts with **different** units.

unit rate: The amount for exactly **one**.

- Pick a benchmark that both amounts **divide** into evenly, like 20 for 2 and 4.
- Scale each rate to the benchmark, then **compare** the results.

Example: Benchmark: **20** minutes for both hoses. Hose A: 6 × 10 = **60** gallons. Hose B: 10 × 5 = **50** gallons. Compare: 60 gallons > 50 gallons in the same 20 minutes. Answer: Hose **A** fills faster.

YOU TRY

1. A: $8 × 4 = $32; B: $10 × 3 = $30. Store B

2. a. 12 (any common multiple) b. 40 (any common multiple)

EXIT TICKET

1. a. 15 miles b. $2.40

2. A: $3 × 3 = $9; B: $5 × 2 = $10. Pack A

3. **A** (divide)

PRACTICE FORM A

1. a. 16 flyers b. 14 stairs c. $2.50 d. 30 miles

2. A: 4 × 15 = 60 tickets; B: 7 × 10 = 70 tickets. Game B

3. Tyler: 140 ÷ 4 = 35; Sarah: 175 ÷ 5 = 35. They type at the same rate.

4. A: $15 × 4 = $60; B: $18 × 3 = $54. Store B

5. a. J: $2.25; K: $2.20 b. J: $27.00; K: $26.40 c. Store K

6. A: $42.50 ÷ 5 = $8.50; B: $66.40 ÷ 8 = $8.30. Rink B

7. a. Reader 1: 40 × 6 = 240; Reader 2: 54 × 5 = 270 b. Reader 2

8. a. A: $120 × 11 = $1,320; B: $176 × 8 = $1,408 b. Company A ($15 vs. $16 per student) c. Sample: it keeps whole numbers, so there is no long division or rounding.

9. Subtracting does not compare rates. Use a 10-ticket benchmark: A = 9 × 5 = 45 points; B = 15 × 2 = 30 points. Ride A gives more points per ticket (4.5 vs. 3).

10. **B** (12)

11. **B** (Store Y)

Self-check word **COMPARE**

PRACTICE FORM B

1. a. 15 flyers b. 12 stairs c. $2.25 d. 32 miles

2. A: 5 × 15 = 75 stickers; B: 8 × 10 = 80 stickers. Machine B

3. Tyler: 156 ÷ 4 = 39; Sarah: 190 ÷ 5 = 38. Tyler

4. A: $9 × 3 = $27; B: $14 × 2 = $28. Store A

5. a. J: $2.50; K: $2.40 b. J: $30.00; K: $28.80 c. Store K

6. A: $34.00 ÷ 4 = $8.50; B: $52.20 ÷ 6 = $8.70. Rink A

7. a. Reader 1: 35 × 6 = 210; Reader 2: 48 × 5 = 240 b. Reader 2

8. a. A: $120 × 11 = $1,320; B: $165 × 8 = $1,320 b. Neither; both cost $15 per student. c. Sample: it keeps whole numbers, so there is no long division or rounding.

9. Subtracting does not compare rates. Use a 15-ticket benchmark: A = 12 × 5 = 60 points; B = 15 × 3 = 45 points. Ride A gives more points per ticket (4 vs. 3).

10. **C** (15)

11. **A** (reach exactly)

Self-check word **BARGAIN**

SHEET 1 · COMPARING RATIOS WITH COMMON TERMS

1. a. 40 b. 42 c. 44 d. 27
2. P yellow: 8, 12, 16, 20; Q yellow: 12, 18, 24, 30. At 15 cups blue, P has 20 yellow and Q has 18 yellow, so Mix Q is bluer.
3. a. 9 : 12 × 4 = 36 : 48 b. 13 : 16 × 3 = 39 : 48 c. Pack B (39 of 48 sprout)
4. No. Scale to 12 spoons sugar: X = 12 : 27, Y = 12 : 30. X has less water for the same sugar, so Drink X is sweeter.

SHEET 2 · COMPARING RATES WITH BENCHMARKS

1. a. 12 b. 18 (any common multiple is correct)
2. Kai: 90 × 3 = 270 words; Lena: 140 × 2 = 280 words. Lena types faster.
3. a. A: $8 × 8 = $64; B: $12 × 5 = $60 b. Store B
4. a. C: $7 × 4 = $28; D: $9 × 3 = $27 b. Store D
5. Benchmark 40 gallons: Car 1 goes 150 × 8 = 1,200 miles; Car 2 goes 200 × 5 = 1,000 miles. Car 1 is more efficient.
6. **B** (24 bagels)
7. **B** (Ella (180 min))

October 2026 conceptual additions — teacher key

Session 1 · Comparing Ratios with Common Terms

A becomes 6:15; B becomes 6:14. B is stronger because it uses less water for the same concentrate. Keep concentrate:water distinct from concentrate:total mixture.

Extension: Use 35 cups water: A 14 cups concentrate, B 15. B remains stronger.

Scoring: 1 point for a valid representation/setup, 1 for accurate mathematics and units, 1 for an explanation connecting the method to the task. Accept equivalent reasoning and suitable oral or diagram responses. Do not infer mastery from confidence alone.

Session 2 · Comparing Rates with Benchmarks

At 10 minutes A fills 40 gallons and B 30. Unit rates are 4 and 3 gallons/minute. A is faster; both methods compare equal times.

Extension: At 20 minutes A fills 80 gallons and B 60; equal scaling preserves the comparison.

Scoring: 1 point for a valid representation/setup, 1 for accurate mathematics and units, 1 for an explanation connecting the method to the task. Accept equivalent reasoning and suitable oral or diagram responses. Do not infer mastery from confidence alone.